WEB PROGRAMMING (InTe3113)

CHAPTER-TWO: HYPER TEXT MARKUP LANGUAGE (HTML)

Compiled by: Kassawmar M.

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HTML

What is HTML?

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages.
- HTML describes the structure of a Web page
- It consists of a series of elements and is a set of markup **tags**
- HTML elements tell the browser how to display the content
- HTML is Not Case Sensitive.

Hypertext Markup Language (HTML)

- An open standard
  - Developed by the World Wide Web Consortium (W3C)
- A language used to format text and objects (such as images) to be displayed in a Web browser.
  - The display language of a browser
- Static—it is only display.
  - Use other technologies to add dynamic function:
    - Client-side (browser) scripting languages (JavaScript)
    - Server-side programs (PHP, Java servlets)

HTML (Cont…)

- The user interface language of the Web.
- An HTML file is a text file containing small markup tags.
  - The markup tags tell the Web browser how to display the page.
    - Appearance
    - Layout
    - Content
- An HTML file must have an htm or html file extension.
- Text mixed with markup tags
  - Tags enclosed in angle brackets

  e.g., <H1>introduction</H1>

HTML (Cont…)

- Constitutes a collection of platform independent styles that define the various components of a web document.
  - Styles indicated by markup tags.
- What is HTML really?
  - Plain-text documents can be created using any text editor.

Why do we use HTML?

- Light
  - Files are flat text and very small
  - Quickly transferred over a network
- Easy to learn
  - Smaller and less complex than other markup languages
  - Less tags (instructions)
- Open standard
  - Works with all platforms, all browsers, and all Web servers
- The rise of Internet technologies

Two parts to HTML

Source code

Website.html - Notepad

File Edit Format View Help

```
<html>
<head>
<title>My website</title>
</head>

<body>
<h1>My website</h1>

<p>Welcome to my website! Click on the links below to
read more about me.</p>

<ul>
<li><a href="website.html">Home</a></li>
<li><a href="family.html">Family</a></li>
<li><a href="photos.html">Photos</a></li>
<li><a href="resume.html">Resume</a></li>
<li><a href="links.html">Links</a></li>
</ul>

</body>
</html>
```

Display

My Website - Microsoft Internet Explorer

File Edit View Favorites Tools Help

My Website

Welcome to my Website! Click on the links below to read more about me.

- Home
- Family
- Photos
- Resume
- Links

Text/HTML editor

HTML files are flat-text files that have a .html (or .htm) extension.

Browser

When you open an HTML file with a browser, the code will run.

HTML Comment

- Comment lines
  - Comments are included between <!--- and --->.
  - Comments cannot be nested.

```
<!--- A comment here --->

<!--- Another comment in

  two lines --->
```

HTML Document Structure

- A HTML document consists of two major portions:
  - Head
    - Contains information about the document, like the title and "meta" data describing the contents.
  - Body
    - Contains the actual matter of the document.
    - Gets displayed within the browser window.

Cont…

```
<!DOCTYPE html>
<html>
  <head>
      (Header: information about the page)
  </head>
  <body>
      (Body:Web page content)
  </body>
</html>
```

Cont…

1. DOCTYPE
2. HTML
3. HEAD
   - TITLE element required
   - Optional elements:
     - BASE
     - META
     - BGSOUND
     - SCRIPT, NOSCRIPT
     - STYLE
     - LINK
4. Body
5. Element

The Basic Web Page structure

```
<html>
    <head>
        <title>  My page </title>
    </head>
    <body>
        Hello this is my page
    </body>
</html>
```

Head section

Body section

Basic HTML syntax

HTML syntax

- Tags
  - Instructions for the browser

    <tag> Some text</tag>
- Nesting
  - Close tags in the opposite order in which you opened them.

    <tag1><tag2>Some text</tag2></tag1>
- Attributes
  - Specify attributes to use non-default behavior

    <tag ***attribute="value">Text</tag>***

HTML Tags

- HTML
  - <HTML>……</HTML>
    - These tags begin and end an HTML document
    - Used to bracket an entire HTML document.
    - Optional for most browsers.
    - Attributes:
      - lang = language code
    - Language code indicates the primary language of the document.
      - en – English

HTML Tags (cont…)

- HEAD
  - <HEAD >……</HEAD>
  - These tags are in the beginning of the document.
  - Important information is stored in-between these tags including: title, meta-data, styles, and programming scripts.
  - Used to provide information about a web page.
  - Nests within itself other tags like

    <base>, <meta> and <title>.

HTML Tags (cont…)

- <base>
  - Attribute: href=url
  - Specifies the base pathname for all relative URLs in the document.
  - No end tag.
- <meta>
  - Used to provide information about a document.
  - Keywords or descriptions to aid search engines.

Cont…

- **Example 2: Codes for HEAD element**

```
<head>
   <title> The title of the Page</title>
   <meta name="Keywords" content="NOUN,Web
     Pages" />
   <meta name="description" content="HTMLBasic Tags"/>
    <base href="http://www.nou.edu.ng " />
     <link rel="stylesheet" type="text/css" href="noun.css"/>
    <script type="text/javascript">
      </script>
      </head>
```

HTML Tags (cont…)

- TITLE
  - <TITLE>……</TITLE>
  - These tags are in-between the HEAD tags and contain the text that appears in the title of the webpage.
  - Usually appears on the title bar of the browser window.
- **Example 3: Code for Title element**

```
<html>
 <head>
      <title>HAWASSA UNIVERSITY</title>
</head>
    <body>     </body>
</html>
```

HTML Tags (cont…)

- BODY
  - <BODY>…..</BODY>
  - As you may have guessed, the body tags contain all the text in the body of the document.
  - Used to bracket the body of a HTML document.
  - Attributes:
    - background=url ⟶ specifies an image file to be used as tiling background.
    - bgcolor=color ⟶ Sets the background color of the document.
    - text=color ⟶ Sets the default color for the normal text in the document.

Cont…

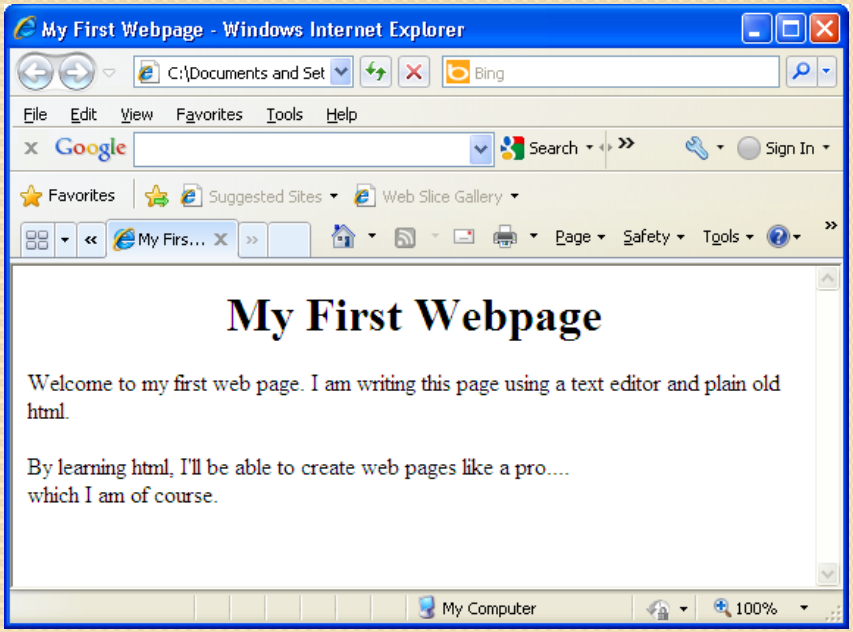
- **Example 4: Codes for Body Element**

```
<html>
    <head>
        <title>Hawassa University Website!
        </title>
  </head>
    <body>
      <h1>Welcome to the official Website of the
            Hawassa University.  </h1>
      </body>
</html>
```

Source code

```
<html>
<head>
<title>My First Webpage</title>
</head>
<body>
<h1 align="center">My First
    Webpage</h1>
<p>Welcome to my first web page. I am
    writing this page using a text editor
    and plain old html.</p>
<p>By learning html, I'll be able to create
    web pages like a pro....<br>
which I am of course.</p>
</body>
</html>
```

display



How to Create and Run HTML codes

- Creating an HTML document is easy.
- To begin coding HTML you need a standard text editor.
- Example: Notepad ,Notepad ++,Visual Studio Code and Dreamweaver are the best editors.
- Steps to use **notepad** application.

➢ **Steps** 1. start → All program → Accessories → notepad

➢ **Steps 2.**Launch Notepad application on your computer

➢ **Steps** 3. Type in your HTML codes

➢ **Steps 4**. Save the document onto a location in your computer drive with a name and the extension "html" (for example IP.html or htm)

➢ **Steps** 5. Click on the filename created.

How to specify colors?

- Two ways:
  - By specifying the red, green, blue or RGB components.
    - Each color encoded in 8 bits.
    - 00 means that the color is turned off; FF means fully turned on.
    - Example:

```
<body text="#FFFFFF"
        bgcolor="#0000FF">
```

How to specify colors?

- By specifying the color name.
  - Some of the colors that are supported by browsers are:

aqua	black	blue	fuchsia
gray	green	lime	maroon
Navy	olive	purple	red
Silver	teal	yellow	white

  - Many other colors are possible.
  - Example:

```
<body text=white>
<body bgcolor="yellow">
```

HTML Backgrounds

- By default, your webpage background is white in color.
- Two good ways to decorate your webpage background.
  - Html Background with Colors
  - Html Background with Images
- The **bgcolor** attribute is used to control the background of an HTML element, specifically page body and table backgrounds
- The **background** attribute used to control the background of an HTML element, specifically page body and table backgrounds.
  - You can specify an image to set background of your HTML page or table.

Text Formatting in HTML

Paragraphs and Headings

- <Hn> ……. </Hn>
  - Used to generate headings, 1 ≤ n ≤ 6.
  - Six different levels of headings.
    - <H1> is the largest, <H6> is the smallest.
- <P>
  - Paragraph marker, used to separate text into paragraphs.
  - End tag </P> is optional.
  - A series of paragraph tags <p><p>…<p> with no intervening text is treated as a single <p>.

Cont..

-
  - Used to indicate that the text following <BR> should begin on the next line. (newline)
- Example:

  This is the first line.

  This is the second line.

  This is the third.

Cont…

- <HR>
  - Produces a horizontal line, which can be used to delimit sections.
  - Length of the line can be specified.
  - Examples:

    <hr>

    <hr size="20"> *[noshade option possible]*

    <hr width="75%">

    <hr align="right" width=120>

    - Across full width of browser, 20 pixels thick, 75% of available page width, and 120 pixels right-justified.

Cont…

- <address> ……. </address>
  - Supplies the contact information of the author.
  - Generally formatted in italics, with a line break above and below.
  - Example

```
<address>
    Mr.  XYZ <br>
    Dept. of Computer Science <br>
    Hawassa University <br>
    Email:  xyz@gmail.com
  </address>
```

Appearance of Text

- …….
  - Displays the enclosed text in bold.
- …….
  - Displays the enclosed text in italics.
- <cite> ……. </cite>
  - Tells the browser that this is a citation. Usually displayed in italics.

Cont…

- <pre> ……. </pre>
  - Allows browser to display carefully laid out text.
  - Used to display program listings.
  - Example:

```
<pre>
   main()
    {
      int i, j;
      abc ();
    }
</pre>
```

Cont…

- _{…….}
  - Displays the enclosed text as subscript.
- ^{…….}
  - Displays the enclosed text as superscript.
- <font> ……. </font>
  - Specifies the style of the enclosed text.
  - Attributes:
    - color = color name
    - face = typeface
    - size = value [1 to 7; 3 is the default]

Cont…

- …….
  - Centers the enclosed elements horizontally on the page.
- <P align=option> ……. </P>
  - Used to align a paragraph.
  - Option can be left, right or center.

Cont…

- **<u>…</u>**
  - To display the elements with underline
- **<strike>.. .</strike>**
  - To displayed with strikethrough, which is a thin line through the text
- **<big>…</big>**
  - element is displayed one font size larger than the rest of the text surrounding it.
- **<small>…</small>** element is displayed one font size smaller

Some Examples

Example 1

```
<html>
    <head>
        <title>My First Webpage</title>
    </head>
    <body text="white" bgcolor="“blue”">
        <h1 align="center">My First Webpage</h1>
        <p>Welcome to my first web page. I am writing this page
        using a text editor and plain old html.</p>
        <p>By learning html, I'll be able to create web pages
        like....<br>
        which I am of course.</p>
    </body>
</html>
```

Example 2

```
<html>
    <head><title>Demonstration of Headings </title></head>
    <body text="#FFFFFF" bgcolor="#0000FF">
            <h1>This is a first level heading. </h1>
            <h2>The second level</h2>
            <h3>The third level</h3>
            <h4>Fourth level. </h4>
            <h5>Fifth level.</h5>
            <h6>And, finally, the sixth .</h6>
            A small amount of plain non-heading text.
    </body>
</html>
```

Example 3

```
<html>
    <head><title>Paragaph Aligning</title></head>
    <body text="white" bgcolor="green">
            <h3>
            <P ALIGN=CENTER> This paragraph will be aligned
                at the center. Even as the browser window
                size changes, the alignment remains the same. </P>
            <P ALIGN=LEFT>This demonstrates left alignment. </P>
            <P ALIGN=RIGHT>How about aligning by the right
                 margin? </P>
            </h3>
    </body>
</html>
```

Example 4

```
<html>
    <head><title>Layout Features 1</title></head>
    <body text="yellow" bgcolor="blue">
            <h2> <pre>
            begin
                if (a > b)
                        then max := a;
                        else max := b;
                end;
            </pre> </h2>
            <hr size=8 width=50%>
            <hr>
            <hr size=20 width="75%" noshade>
    </body>
</html>
```

Example 5

```
<html>
   <head><title>Layout Features 2</title></head>
   <body text="yellow" bgcolor="blue">
         <h3>Extended Quotations</h3>
            <blockquote>
                  <P>This is the first paragraph within
                              the BLOCKQUOTE environment. </P>
                  <P>Another paragraph starts here. </P>
                              We type some text here without explicitly
                              giving paragraph break.
         </blockquote>
   </body>
</html>
```

Example 6

```
<html>
    <head><title> Superscript and Subscript </title></head>
    <body text="white" bgcolor="blue">
            <h1> y = x <sup> 3 </sup> + 2 x <sup> 2 </sup> + 4 </h1>
            <br>
            <h2> W <sub> total </sub> = x <sup> 2 </sup> - 5 <h2>
    </body>
</html>
```